MATHS MANTHAN
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CBSE CLASS 10 | MATHEMATICS

SURFACE AREAS AND VOLUMES

Master Concept, Formula & Visual Problem-Solving Handbook

Academic Year 2026–27

See the solid. Identify the exposed surface.
Calculate with confidence.

Core formulas | Visual diagrams | Solved examples | Exam checks

A student-friendly Maths Manthan revision resource



What this handbook includes

- Formulae for cubes, cuboids, cylinders, cones, spheres and hemispheres
- Surface area and volume of combined solids
- Visual diagrams, fully worked examples and a quick formula matrix

How to use it

Identify every dimension first. Decide which surfaces remain exposed, keep units consistent, write the formula before substitution and check the final units.

Official syllabus scope

The CBSE Class X Mathematics syllabus for 2026–27 lists combinations of any two of these solid types: cubes, cuboids, spheres, hemispheres and right circular cylinders/cones. This handbook labels hollow solids, cavities, density, displacement and percentage-volume models as **EXTENSION PRACTICE**; they are useful applications, but are not all explicitly listed in that syllabus statement. See the official CBSE curriculum and follow your school's current instructions.

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Prepared for learning and revision – not an official CBSE publication.

CHAPTER 12

SURFACE AREAS AND VOLUMES

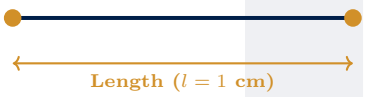
CBSE CLASS 10 · 2026–27

Concepts, Formulae and Visual Problem-Solving Handbook

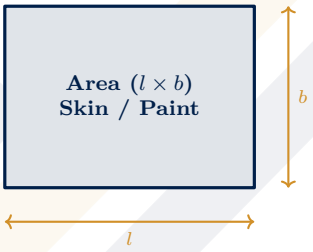
Module 1 | Prerequisite Dimensional Thinking (1D vs 2D vs 3D)

Before applying spatial formulas, you must build absolute clarity regarding the physical dimensions:

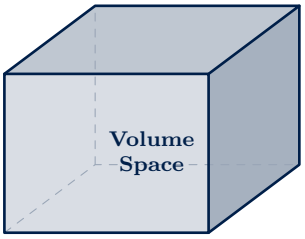
1D: Linear Measure
Perimeter, Edge, Slant Height
Unit: cm, m, mm



2D: Surface Area
CSA, TSA, Base Area
Unit: cm^2, m^2



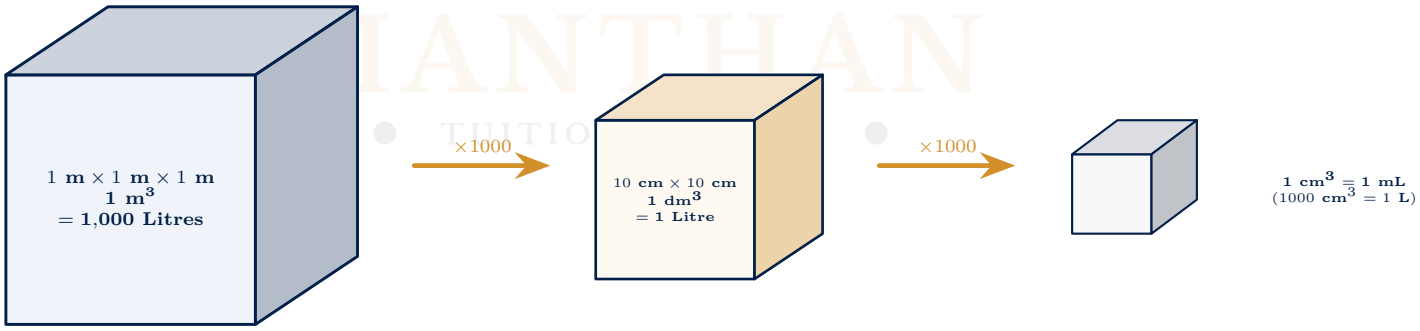
3D: Capacity / Volume
Space occupied; liquid capacity
Unit: cm^3, m^3, L



The Mensuration Golden Rule: In any multi-step question, never mix different units! Always convert all linear dimensions (r, h, l, a, b) into the **same unit** (cm or m) before substituting them into any surface area or volume formula.

Module 2 | The Metric Volume & Capacity Conversion Matrix

Practical applications often use liquid capacity and material density. These conversions connect cubic units with litres and millilitres:



Fundamental Volume-to-Capacity Equations for Board Exams:

- 1 m^3 = 1,000,000 cm^3 = 10^6 cm^3
- 1 m^3 = 1,000 dm^3 = **1,000** Litres = 1 kilolitre (kL)
- 1 Litre = **1,000** cm^3 = 1 dm^3
- 1 cm^3 = **1** mL = $\frac{1}{1000}$ Litre = 10^{-3} Litres

Module 3 | The Value of π Standard Conventions

Choosing a value for π:

1. Follow the value of π specified in the question.
2. If none is specified, keep π exact when practical; otherwise use a suitable approximation such as $\frac{22}{7}$ or 3.14.
3. Keep π symbolic while simplifying, then substitute a numerical value at the end if needed.

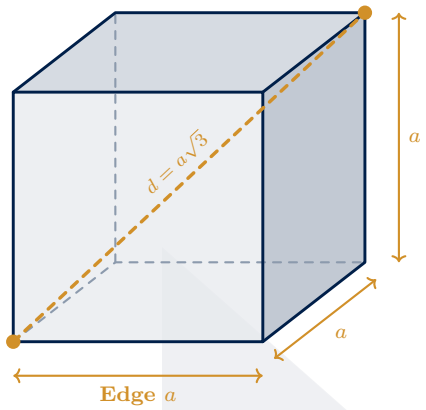
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SOLID 1 & 2 | Prismatic Polyhedra: The Cube and The Cuboid

CORE SYLLABUS

Solid 1 | The Cube (Regular Hexahedron)

A cube is a 3-dimensional solid bounded by six congruent square faces. All twelve edges are of equal length a.

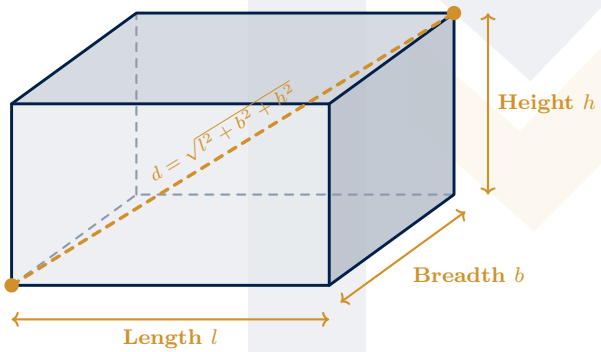


Cube Formula Suite (Edge length = a):

- **Volume (V):** $\mathbf{V = a^3}$
- **Lateral Surface Area (LSA):** LSA $= \mathbf{4a^2}$ (4 walls)
- **Total Surface Area (TSA):** TSA $= \mathbf{6a^2}$ (all 6 faces)
- **Face Diagonal (d_f):** $d_f = a\sqrt{2}$
- **Space Diagonal / Longest Rod (d):** $\mathbf{d = a\sqrt{3}}$
- **Total Edge Length:** $12a$

Solid 2 | The Cuboid (Rectangular Parallelepiped)

A cuboid has six rectangular faces. Opposite faces are identical. The dimensions are length l, breadth b, and height h.



Cuboid Formula Suite (l, b, h):

- **Volume (V):** $\mathbf{V = l \cdot b \cdot h}$ = Base Area $\times\, h$
- **Area of 4 Walls (LSA):** LSA $= \mathbf{2h(l + b)}$
- **Total Surface Area (TSA):** TSA $= \mathbf{2(lb + bh + hl)}$
- **Open Box (5 Faces):** Area $= 2h(l+b)+lb =$ TSA $- lb$
- **Space Diagonal (Longest Rod):** $\mathbf{d = \sqrt{l^2 + b^2 + h^2}}$
- **Total Edge Length:** $4(l + b + h)$

Exam Trap Alert | Longest Rod in a Room: Always compute the 3D space diagonal $d = \sqrt{l^2 + b^2 + h^2}$, *never* the 2D floor diagonal $\sqrt{l^2 + b^2}$!

Quick worked check

Cube with edge $a = 4$ cm

$V = a^3 = 64\,\text{cm}^3$
LSA $= 4a^2 = 64\,\text{cm}^2$
TSA $= 6a^2 = 96\,\text{cm}^2$
$d = a\sqrt{3} = 4\sqrt{3}\,\text{cm}$

Cuboid with $l = 8$, $b = 5$, $h = 4$ cm

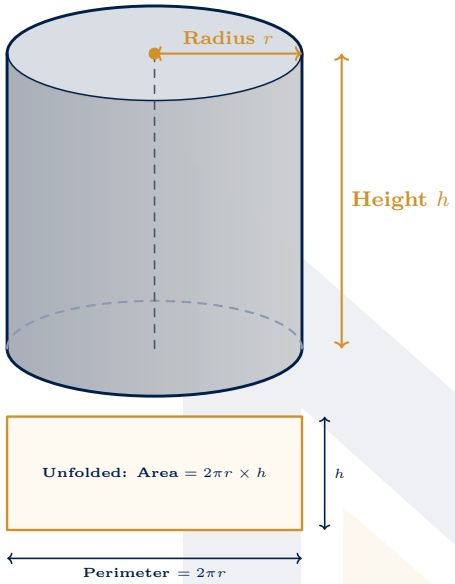
$V = lbh = 160\,\text{cm}^3$
TSA $= 2(lb + bh + hl) = 184\,\text{cm}^2$
4 walls $= 2h(l + b) = 104\,\text{cm}^2$
$d = \sqrt{l^2 + b^2 + h^2} = \sqrt{105}\,\text{cm}$

SOLID 3 | Circular Prisms: Solid Cylinder & Hollow Metallic Pipe

CORE SYLLABUS

Solid 3A | Solid Right Circular Cylinder

Generated by rotating a rectangle about one of its sides. It has two parallel flat circular bases of radius r and one smooth curved surface of vertical height h.



Solid Cylinder Formula Suite (r, h):

- **Curved Surface Area (CSA):**

$$\text{CSA} = 2\pi rh$$

(Physical meaning: Area when cylindrical label is unrolled into a flat rectangle).

- **Area of Both Circular Bases:**

$$2 \times (\pi r^2) = 2\pi r^2$$

- **Total Surface Area (TSA):**

$$\text{TSA} = \text{CSA} + 2(\text{Base Area}) = 2\pi rh + 2\pi r^2 = \mathbf{2\pi r(r + h)}$$

- **Surface Area of Open Cylinder (e.g., Cup):**

$$\text{Area} = 2\pi rh + \pi r^2 = \pi\mathbf{r(2h + r)}$$

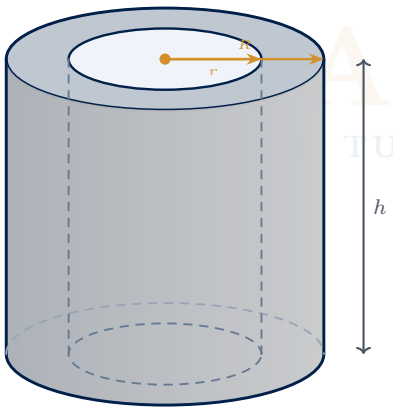
- **Volume (V):**

$$V = \text{Base Area} \times \text{Height} = \pi\mathbf{r^2 h}$$

Solid 3B | Hollow Cylinder (Cylindrical Metallic Shell / Pipe)

EXTENSION PRACTICE

A hollow cylinder has an inner radius r, outer radius R, wall thickness $t = R - r$, and height h. It has two cylindrical surfaces (inside and outside) and two ring-shaped circular faces (ends).



Hollow Cylinder Formula Suite (R, r, h):

- **Wall Thickness (t):** $t = R - r$
- **Area of Each Annular End Face:**

$$\text{Area}_{\text{ring}} = \pi(R^2 - r^2)$$

- **Internal Curved Surface Area:** $\text{CSA}_{\text{in}} = 2\pi rh$
- **External Curved Surface Area:** $\text{CSA}_{\text{out}} = 2\pi Rh$
- **Total Surface Area (Both ends open pipe):**

$$\text{TSA} = \text{CSA}_{\text{out}} + \text{CSA}_{\text{in}} + 2(\text{Area}_{\text{ring}})$$

$$= \mathbf{2\pi Rh} + \mathbf{2\pi rh} + \mathbf{2\pi(R^2 - r^2)}$$

$$= \mathbf{2\pi(R + r)(h + R - r)}$$

- **Volume of Material / Metal (V_{metal}):**

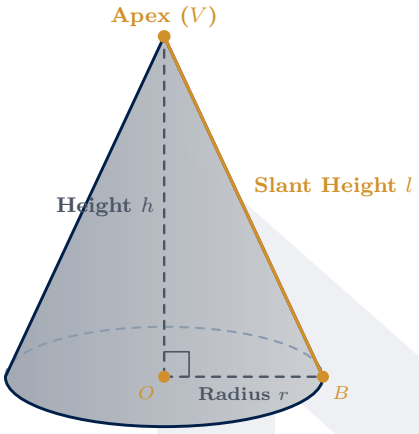
$$\mathbf{V} = \pi(\mathbf{R^2 - r^2})\mathbf{h} = \pi\mathbf{h(R + r)(R - r)}$$

SOLID 4 | Circular Pyramids: The Right Circular Cone

CORE SYLLABUS

Solid 4 | Geometry & The Fundamental Right Triangle Relation

A right circular cone is generated by rotating a right-angled triangle about one of its perpendicular legs. Its vertical axis h meets the circular base center perpendicularly at O.



Pythagorean Slant Height Theorem in ΔVOB:

$$l^2 = h^2 + r^2$$

- **Slant Height (l):**

$$\mathbf{l = \sqrt{h^2 + r^2}}$$

- **Vertical Height (h):**

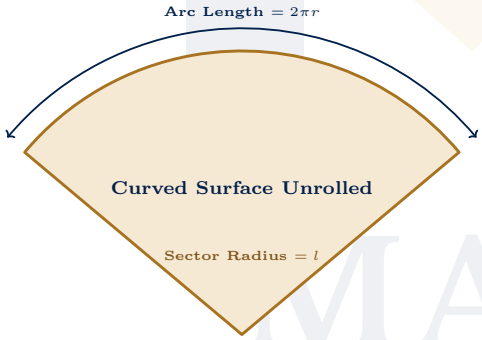
$$\mathbf{h = \sqrt{l^2 - r^2}}$$

- **Base Radius (r):**

$$\mathbf{r = \sqrt{l^2 - h^2}}$$

Always calculate slant height l first before evaluating the curved surface area!

Cone Surface Area & Volume Formulas



Archimedes' Volume Ratio:
1 Cylinder $= 3\times$ Cones of same (r, h)

$\mathbf{V_{cone} = \frac{1}{3} V_{cylinder}}$

Cone Formula Suite (r, h, l):

- **Curved Surface Area (CSA):**

$$\text{CSA} = \pi\mathbf{rl}$$

(Area of the unrolled circular sector of radius l and arc length $2\pi r$).

- **Total Surface Area (TSA):**

$$\text{TSA} = \text{CSA} + \text{Base Area} = \pi rl + \pi r^2 = \pi\mathbf{r(l + r)}$$

- **Volume (V):**

$$\mathbf{V = \frac{1}{3}\pi r^2 h}$$

(Exactly one-third of a cylinder with identical base radius and vertical height).

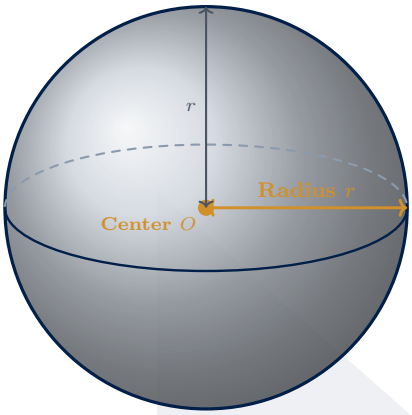
Trap Alert | Slant Height (l) vs. Vertical Height (h):
In formula CSA $= \pi rl$, you **must** use the slant height l. Never substitute vertical height h in place of l. Conversely, in volume $V = \frac{1}{3}\pi r^2 h$, you **must** use vertical height h, never l.

SOLID 5 | Spherical Solids: Sphere, Solid Hemisphere & Hollow Bowl

CORE SYLLABUS

Solid 5A | The Complete Sphere

A sphere is the set of all points in 3-dimensional space that are equidistant from a fixed center point O. It has only one continuous curved surface (no edges, no flat bases).



Sphere Formula Suite (Radius = r):

- **Surface Area (S):**

$$\mathbf{S = 4\pi r^2}$$

(A sphere has only one surface: CSA = TSA = $4\pi r^2$, equal to the area of exactly 4 flat circles of radius r).

- **Volume (V):**

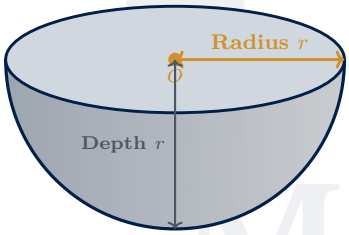
$$\mathbf{V = \frac{4}{3}\pi r^3}$$

- **Circumference of Great Circle:**

$$C = 2\pi r$$

Solid 5B | Solid Hemisphere (Half of a Sphere)

Formed by slicing a solid sphere through its equator plane. It features **two distinct surfaces**: a curved dome and a flat circular base.



Solid Hemisphere Formula Suite (r):

- **Curved Surface Area (CSA):**

$$\text{CSA} = \mathbf{2\pi r^2} \quad \text{(Dome only)}$$

- **Total Surface Area (TSA):**

$$\text{TSA} = \text{Curved Dome} + \text{Flat Circular Base}$$

$$\text{TSA} = \mathbf{2\pi r^2} + \pi\mathbf{r^2} = \mathbf{3\pi r^2}$$

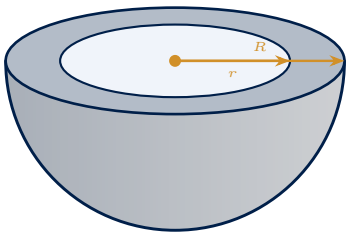
- **Volume (V):**

$$\mathbf{V = \frac{2}{3}\pi r^3} \quad \text{(Exactly half of sphere)}$$

Solid 5C | Hollow Hemispherical Bowl (With Thickness)

EXTENSION PRACTICE

A hemispherical bowl with inner radius r, outer radius R, and wall thickness $t = R - r$.



Hollow Hemispherical Bowl (R, r):

- **Outer CSA:** $2\pi R^2$ | **Inner CSA:** $2\pi r^2$
- **Area of Circular Rim:** $\pi(R^2 - r^2)$
- **Total Surface Area (TSA):**

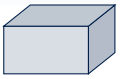
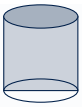
$$\text{TSA} = 2\pi R^2 + 2\pi r^2 + \pi(R^2 - r^2) = \pi(\mathbf{3R^2} + \mathbf{r^2})$$

- **Volume of Material:** $\mathbf{V = \frac{2}{3}\pi(R^3 - r^3)}$

MASTER REFERENCE MATRIX | Core Solids and Hollow-Cylinder Extension

QUICK REFERENCE

Use this master comparative lookup grid to cross-verify all standard formulas during problem-solving:

Solid Name	3D Diagram	CSA / LSA	TSA	Volume (V)	Key Relationships
Cube Edge $= a$		$4a^2$	$6a^2$	a^3	Diagonal $d = a\sqrt{3}$ Edges $= 12a$
Cuboid l, b, h		$2h(l+b)$	$2(lb+bh+hl)$	$l \cdot b \cdot h$	Diagonal: $\sqrt{l^2+b^2+h^2}$ 4 walls $= 2h(l+b)$
Cylinder Radius r, Ht h		$2\pi rh$	$2\pi r(r+h)$	$\pi r^2 h$	Open: $\pi r(2h+r)$ Base Area $= \pi r^2$
Hollow Cylinder Radii R, r, Ht h		$2\pi Rh$ (out) $2\pi rh$ (in)	$2\pi(R+r)$ $\times(h+R-r)$	$\pi(R^2-r^2)h$	Thickness $t = R - r$ Ring: $\pi(R^2-r^2)$
Right Cone r, h, l		πrl	$\pi r(l+r)$	$\frac{1}{3}\pi r^2 h$	$l = \sqrt{r^2+h^2}$ $V_{\text{cyl}} = 3V_{\text{cone}}$
Sphere Radius r		$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$	Only 1 surface! CSA $=$ TSA $=$ $4\pi r^2$
Hemisphere Radius r		$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$	Dome $= 2\pi r^2$ Base $= \pi r^2$

Master Volume Ratio Triplet (Same base radius r and common height $h = r$):
For a cone, hemisphere, and cylinder each having base radius r and height $h = r$:

$$V_{\text{cone}} : V_{\text{hemisphere}} : V_{\text{cylinder}} = \frac{1}{3}\pi r^3 : \frac{2}{3}\pi r^3 : \pi r^3 = \mathbf{1 : 2 : 3}$$

The ratio follows directly by substituting the same radius and height into each volume formula.

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WORKED EXAMPLES | From Formula to Final Answer

WORKED PRACTICE

Example 1: Cylinder - surface area and volume

A cylinder has radius $r = 7\,\text{cm}$ and height $h = 10\,\text{cm}$. Find its curved surface area, total surface area and volume. Take $\pi = \frac{22}{7}$.

$$\text{CSA} = 2\pi rh = 2 \times \frac{22}{7} \times 7 \times 10 = \boxed{440\,\text{cm}^2},$$

$$\text{TSA} = 2\pi r(r+h) = 2 \times \frac{22}{7} \times 7 \times (7+10)$$

$$= \boxed{748\,\text{cm}^2},$$

$$V = \pi r^2 h = \frac{22}{7} \times 7^2 \times 10 = \boxed{1540\,\text{cm}^3}.$$

Check: CSA and TSA are measured in square centimetres; volume is measured in cubic centimetres.

Example 2: Cone - slant height, area and volume

A cone has radius $r = 7\,\text{cm}$ and vertical height $h = 24\,\text{cm}$. Find the slant height, CSA, TSA and volume.

$$l = \sqrt{r^2 + h^2} = \sqrt{7^2 + 24^2} = \sqrt{625} = 25\,\text{cm},$$

$$\text{CSA} = \pi rl = \frac{22}{7} \times 7 \times 25 = \boxed{550\,\text{cm}^2},$$

$$\text{TSA} = \pi r(l+r) = \frac{22}{7} \times 7 \times (25+7) = \boxed{704\,\text{cm}^2},$$

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 49 \times 24 = \boxed{1232\,\text{cm}^3}.$$

Remember: CSA uses slant height l; volume uses vertical height h.

Example 3: Cone + hemisphere toy

A toy is made by joining a cone to a hemisphere of the same radius $r = 3\,\text{cm}$. The cone's vertical height is $4\,\text{cm}$, so its slant height is $l = 5\,\text{cm}$. Find the outside surface area and total volume.

$$\text{Outside area} = \pi rl + 2\pi r^2 = 15\pi + 18\pi = \boxed{33\pi\,\text{cm}^2},$$

$$V = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$$

$$= 12\pi + 18\pi = \boxed{30\pi\,\text{cm}^3}.$$

The joined circular base is hidden, so it is not included in the outside surface area.

Quick practice | Answers for self-check

No.	Question	Answer
1	Cuboid $l = 8$, $b = 5$, $h = 4$ cm: find TSA and volume.	$184\,\text{cm}^2$; $160\,\text{cm}^3$
2	Sphere with radius 3 cm: find surface area and volume.	$36\pi\,\text{cm}^2$; $36\pi\,\text{cm}^3$
3	Tent has cylindrical wall $r = 7$ cm, $h = 10$ cm and cone roof slant height 25 cm. Find canvas area (no floor).	$315\pi\,\text{cm}^2 = 990\,\text{cm}^2$ for $\pi = 22/7$

COMBINATIONS OF SOLIDS | The Non-Additive Surface Area Law

EXAM STRATEGY

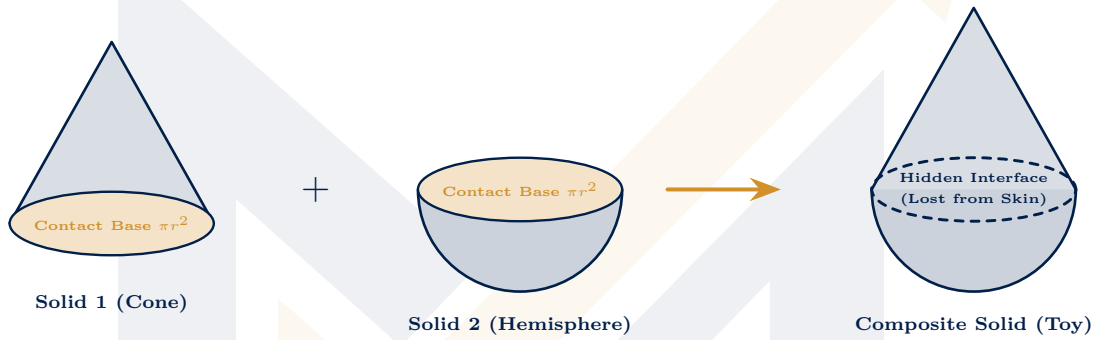
The CBSE Class X 2026–27 syllabus includes surface areas and volumes of combinations of any two listed solid types. When two non-overlapping solids are joined along matching faces, use these two foundational laws:

Law 1 | The Non-Additive Surface Area Principle

CRITICAL AXIOM: When two solids are joined face-to-face, the Total Surface Area of the resulting composite solid is **NEVER** equal to the sum of their individual total surface areas:

$$\text{TSA}_{\text{composite}} \neq \text{TSA}_{\mathbf{1}} + \text{TSA}_{\mathbf{2}}$$

Because the contact faces are glued together and concealed inside the interior, they no longer form part of the outer skin!



The Universal Composite Surface Area Equation:

$$\text{TSA}_{\text{composite}} = \text{Visible Area of Solid 1} + \text{Visible Area of Solid 2}$$

Or equivalently: $\text{TSA}_{\text{composite}} = \text{TSA}_{\mathbf{1}} + \text{TSA}_{\mathbf{2}} - \mathbf{2} \times (\text{Contact Base Area})$

For the cone-hemisphere toy:

$$\text{TSA}_{\text{toy}} = \text{CSA}_{\text{cone}} + \text{CSA}_{\text{hemisphere}} = \pi\mathbf{rl} + \mathbf{2}\pi\mathbf{r^2}$$

Law 2 | The Strict Additive Volume Principle

For two non-overlapping solids joined along a common face, volume is additive: their volumes sum because the shared boundary has zero volume:

The Universal Composite Volume Equation:

$$\mathbf{V}_{\text{composite}} = \mathbf{V_1} + \mathbf{V_2}$$

For the cone-hemisphere toy:

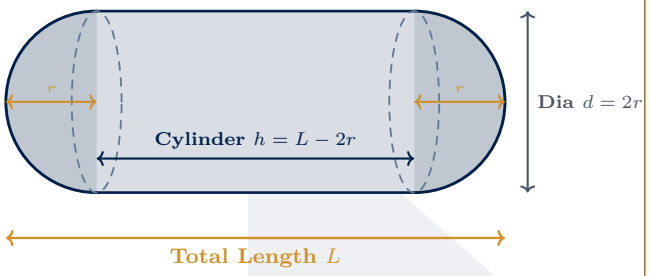
$$\mathbf{V}_{\text{toy}} = \mathbf{V}_{\text{cone}} + \mathbf{V}_{\text{hemisphere}} = \frac{\mathbf{1}}{\mathbf{3}}\pi\mathbf{r^2h} + \frac{\mathbf{2}}{\mathbf{3}}\pi\mathbf{r^3} = \frac{\mathbf{1}}{\mathbf{3}}\pi\mathbf{r^2}(\mathbf{h} + \mathbf{2r})$$

EXAM ARCHETYPES I | The Medicine Capsule & The Spinning Top

CORE SYLLABUS

Model 1 | The Medicine Capsule / Storage Tanker

A solid composed of a central cylinder with two hemispherical caps attached to each end.



Capsule Formula Suite:

- **Radius:** $r = d/2$
- **Cylindrical Height (h):**

$$\mathbf{h = L - 2r}$$

- **Total Surface Area (TSA):**

$$\text{TSA} = \text{CSA}_{\text{cyl}} + 2 \times \text{CSA}_{\text{hemi}}$$

$$= 2\pi rh + 2(2\pi r^2) = 2\pi rh + 4\pi r^2$$

$$\text{TSA} = \mathbf{2\pi r(h + 2r)}$$

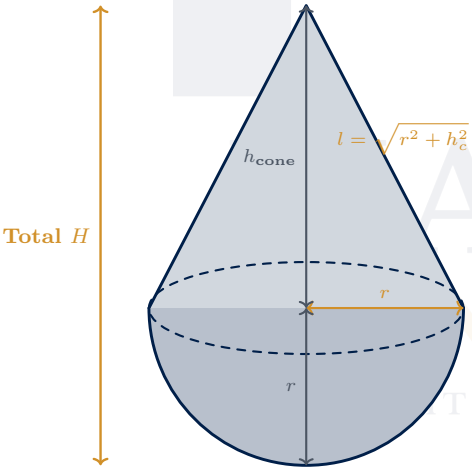
- **Total Volume (V):**

$$V = V_{\text{cyl}} + 2 \times V_{\text{hemi}} = \pi r^2 h + \frac{4}{3}\pi r^3$$

$$\mathbf{V = \pi r^2 \left(h + \frac{4}{3}r\right)}$$

Model 2 | The Toy / Spinning Top (Lattu)

A toy formed by mounting a right circular cone on top of a hemisphere of identical radius r.



Toy Formula Suite:

- **Cone Height:** $h_{\text{cone}} = H - r$
- **Cone Slant Height:** $l = \sqrt{r^2 + h_{\text{cone}}^2}$
- **Total Surface Area (TSA):**

$$\text{TSA} = \text{CSA}_{\text{cone}} + \text{CSA}_{\text{hemi}}$$

$$\text{TSA} = \pi \mathbf{rl} + \mathbf{2\pi r^2} = \pi \mathbf{r(l + 2r)}$$

- **Total Volume (V):**

$$V = V_{\text{cone}} + V_{\text{hemi}} = \frac{1}{3}\pi r^2 h_{\text{cone}} + \frac{2}{3}\pi r^3$$

$$\mathbf{V = \frac{1}{3}\pi r^2 (h_{\text{cone}} + 2r)}$$

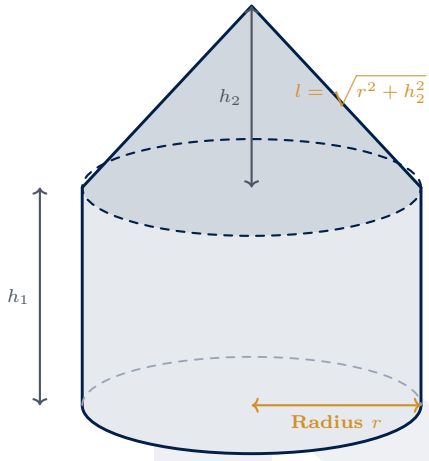
Crucial Board Note: When calculating the surface area of a capsule or toy, **never** add πr^2 for the base! The base circle is completely hidden inside the solid and cannot be painted, polished, or touched.

EXAM ARCHETYPES II | Circus Tent / Silo & End-to-End Cubes

CORE SYLLABUS

Model 3 | The Circus Tent / Grain Silo (Cylinder + Conical Roof)

A cylindrical structure of radius r and height h_1, topped by a conical roof of the same radius r, vertical height h_2, and slant height l.



Circus Tent Formula Suite:

- **Roof Slant Height:** $l = \sqrt{r^2 + h_2^2}$
- **Canvas Required for Tent:** *(Note: The ground base is open floor, not canvas)*

$$\text{Area} = \text{CSA}_{\text{cylinder}} + \text{CSA}_{\text{cone}}$$

$$\text{Area} = \mathbf{2\pi r h_1 + \pi r l = \pi r(2h_1 + l)}$$

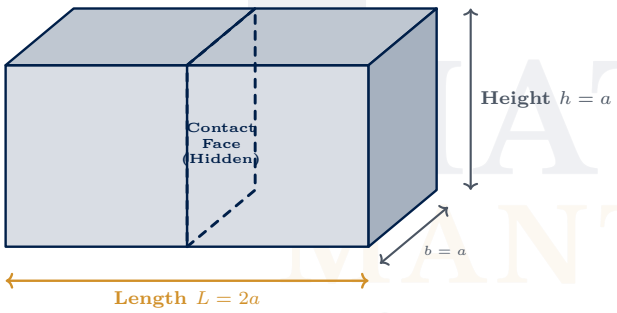
- **Cost of Canvas:** Area $\times$ Rate per m^2
- **Air Capacity / Total Volume (V):**

$$V = V_{\text{cylinder}} + V_{\text{cone}} = \pi r^2 h_1 + \frac{1}{3}\pi r^2 h_2$$

$$\mathbf{V = \pi r^2 \left(h_1 + \frac{1}{3}h_2\right)}$$

Model 4 | Two or More Cubes Joined End-to-End

When n identical cubes of edge length a are joined face-to-face in a straight row, they fuse into a single cuboid.



End-to-End Cubes Joined:

- **Resulting Dimensions:**

$$\text{Length } L = 2a, \quad \text{Breadth } b = a, \quad \text{Height } h = a$$

- **Total Surface Area (TSA):**

$$\text{TSA} = 2(Lb + bh + hL)$$

$$= 2(2a \cdot a + a \cdot a + a \cdot 2a)$$

$$= 2(2a^2 + a^2 + 2a^2) = 2(5a^2) = \mathbf{10a^2}$$

- **Contact Loss Check:**

$$\text{Sum of separate TSAs} = 6a^2 + 6a^2 = 12a^2$$

$$\text{Contact area lost} = 2 \times a^2 = 2a^2$$

$$\text{Final TSA} = 12a^2 - 2a^2 = \mathbf{10a^2}$$

- **For n Cubes in a Row:** $\text{TSA} = \mathbf{(4n + 2)a^2}$

SCOOPED-OUT SOLIDS | Cavity Formation & Surface-Area Accounting

EXTENSION PRACTICE

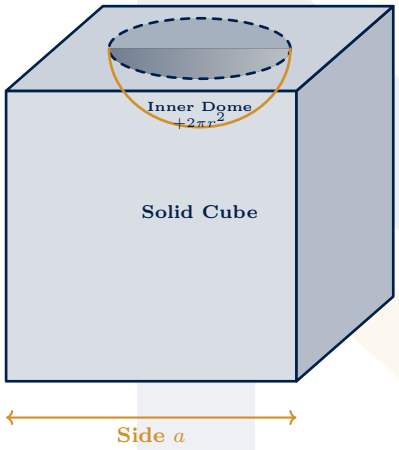
Cavity Surface-Area Accounting

THE CAVITY PARADOX: When a depression or cavity is scooped out of a solid block:

- **VOLUME DECREASES:** Material has been removed: $V_{\text{remaining}} = V_{\text{original}} - V_{\text{cavity}}$.
- **A NEW INTERIOR SURFACE IS EXPOSED:** The cavity adds an internal surface, but the opening area is removed from the original outer surface. Compute the net change rather than assuming that total surface area always increases.

Model 5 | Cube with a Hemispherical Depression Scooped Out

A hemispherical cavity of diameter $d \leq a$ (radius $r \leq a/2$) is hollowed out from the top face of a wooden cube of side a.



Cube with Hemispherical Cavity (a, r):

- **Total Surface Area (TSA):**

$$\text{TSA} = \text{TSA of Cube} - \text{Base Circle} + \text{Inner Dome}$$

$$\text{TSA} = 6a^2 - \pi r^2 + 2\pi r^2$$

$$\text{TSA} = \mathbf{6a^2 + \pi r^2}$$

(Identical formula whether the hemisphere is scooped out as a cavity OR mounted on top as a dome!)

- **For Maximum Diameter $(d = a \implies r = a/2)$:**

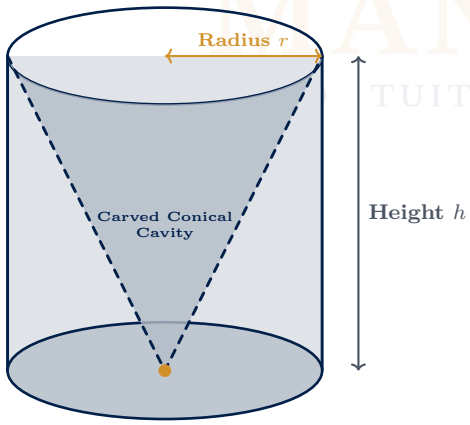
$$\text{TSA} = \mathbf{6a^2} + \pi\left(\frac{\mathbf{a}}{\mathbf{2}}\right)^{\mathbf{2}} = \mathbf{a^2}\left(\mathbf{6} + \frac{\pi}{4}\right)$$

- **Remaining Volume (V):**

$$\mathbf{V}_{\text{remaining}} = \mathbf{a^3} - \frac{\mathbf{2}}{\mathbf{3}}\pi\mathbf{r^3}$$

Model 6 | Solid Cylinder with a Conical Cavity

A solid right circular cylinder of height h and radius r from which a conical cavity of identical height h and base radius r is drilled out.



Cylinder with Conical Cavity (r, h):

- **Total Surface Area of Remaining Solid:**

$$\text{TSA} = \text{CSA}_{\text{cyl}} + \text{Base Area} + \text{CSA}_{\text{cone}}$$

$$= \mathbf{2\pi rh} + \pi\mathbf{r^2} + \pi\mathbf{rl} = \pi\mathbf{r}(\mathbf{2h} + \mathbf{r} + \mathbf{l})$$

where $l = \sqrt{r^2 + h^2}$.

- **Volume of Remaining Solid:**

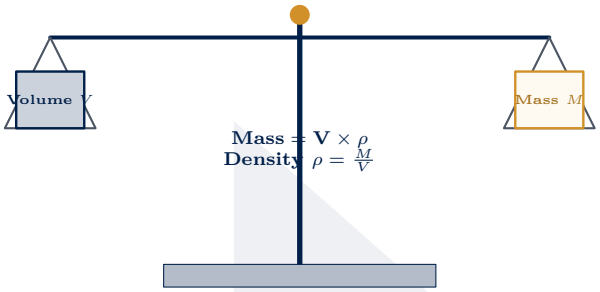
$$V_{\text{remaining}} = V_{\text{cylinder}} - V_{\text{cone}} = \pi r^2 h - \frac{1}{3}\pi r^2 h$$

$$\mathbf{V}_{\text{remaining}} = \frac{\mathbf{2}}{\mathbf{3}}\pi\mathbf{r^2h}$$

EXTENSION PRACTICE

Module 11.1 | Physical Mass and Density Calculations

In applied problems, density can be used to calculate the mass of material in a pipe, pillar or sphere. A density of $7.8\,\text{g/cm}^3$ means each cubic centimetre has a mass of 7.8 g. Mass and weight are different physical quantities.



Mass, Volume & Density Laws:

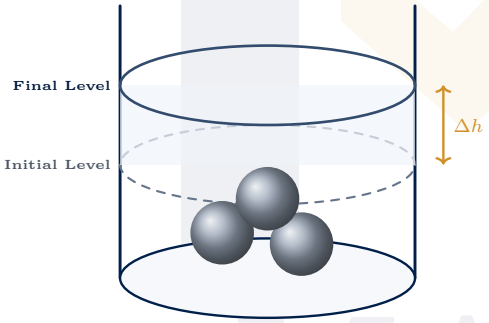
- **Fundamental Mass Formula:**

$$\text{Mass} = \text{Volume of Material} \times \text{Density}$$

- **Unit Consistency Check:**
  - If volume is in cm^3 and density in g/cm^3, mass is obtained in **grams (g)**.
  - To convert grams to kilograms: $\text{Mass in kg} = \frac{\text{Mass in g}}{1000}$.
- **For Hollow Metallic Pipe:**

$$\text{Mass} = [\pi(\mathbf{R^2} - \mathbf{r^2})\mathbf{h}] \times \text{Density}$$

Module 11.2 | Volume Displaced & The Immersion Principle



Liquid Displacement Theorem:

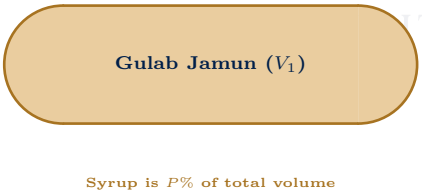
$$\mathbf{V}_{\text{submerged solids}} = \mathbf{V}_{\text{liquid displaced}}$$

When n spherical lead shots of radius r_{sphere} are dropped into a cylindrical vessel of radius R:

$$\mathbf{n} \times \left(\frac{\mathbf{4}}{\mathbf{3}}\pi\mathbf{r}^{\mathbf{3}}_{\text{sphere}}\right) = \pi\mathbf{R^2\Delta h}$$

$$\mathbf{n} = \frac{\pi\mathbf{R^2\Delta h}}{\frac{\mathbf{4}}{\mathbf{3}}\pi\mathbf{r}^{\mathbf{3}}_{\text{sphere}}} = \frac{\mathbf{3R^2\Delta h}}{\mathbf{4r}^{\mathbf{3}}_{\text{sphere}}}$$

Module 11.3 | Percentage Sugar Syrup (The Gulab Jamun Model)



Simplified Percentage-by-Volume Model for N Sweets:

1. Find volume of 1 sweet: $V_1 = \pi r^2 h_{\text{cyl}} + \frac{4}{3}\pi r^3$.
2. Total volume of N sweets: $V_{\text{total}} = N \times V_1$.
3. If the question defines syrup volume as P% of total sweet volume:

$$\mathbf{V}_{\text{syrup}} = \left(\frac{\mathbf{P}}{\mathbf{100}}\right) \times \mathbf{N} \times \mathbf{V_1}$$

BOARD MASTERY | Common Traps, Smart Factoring & Final Checklist

EXAM STRATEGY

Module 12.1 | The Top 6 Board Examination Blunders (And How to Bust Them)

Trap 1: The Diameter vs. Radius Trap: In "Diameter 14 cm", substituting $r = 14$.

TRAP ALERT

Always write $\mathbf{r = d/2 = 7}$ cm first!

Trap 2: The Contact Interface Addition Trap: Adding base circles that are joined together. Contact faces are **hidden inside**; TSA includes only exposed boundaries!

Trap 3: The Cone Height Confusion (h vs. l): Using h in CSA $= \pi rl$. Use slant height l in the cone CSA formula; use vertical height h in its volume formula.

Trap 4: The Mixed Units Catastrophe: Multiplying meters with centimeters without conversion. Unify all dimensions to cm or m before calculating!

Trap 5: The Cavity Surface-Area Trap: Do not simply subtract the cavity area. Subtract the removed opening area and add the newly exposed inner surface. For a hemispherical depression cut into a cube, the net change is $+\pi r^2$, so TSA $= 6a^2 + \pi r^2$.

Trap 6: Premature Decimal Clutter: Multiplying $\pi = \frac{22}{7}$ in early steps. Keep π as a symbol and factor it out until the final line.

Module 12.2 | The High-Speed Algebraic Factorization Method

Compare the traditional tedious calculation versus the professional Maths Manthan method:

The Slow Method (High Arithmetic Risk)	The Maths Manthan Factoring Method
Find TSA of toy ($r = 3.5, l = 12.5$): TSA $= \frac{22}{7} \times 3.5 \times 12.5 + 2 \times \frac{22}{7} \times 3.5^2$ $= \frac{22}{7} \times 43.75 + \frac{22}{7} \times 24.5$ $= 137.5 + 77.0 = 214.5$ cm^2	Factor common algebraic terms first: TSA $= \pi rl + 2\pi r^2 = \pi\mathbf{r(l + 2r)}$ Substitute once at the end: $= \frac{22}{7} \times 3.5 \times (12.5 + 7) = 11 \times 19.5 = \mathbf{214.5}$ cm$^\mathbf{2}$

Module 12.3 | The 5-Point Board Examination Pre-Submission Checklist

Final 60-Second Answer Sheet Audit:

- ☐ **Given Data Stated:** Explicitly state radius ($r = d/2$), heights, and unified dimensions.
- ☐ **Formula Mentioned:** State the symbolic formula before substituting values; this makes your method clear and can support method marks.
- ☐ **Slant Height Verified:** If a cone is involved, show $l = \sqrt{r^2 + h^2}$ clearly.
- ☐ **Correct π Value:** Follow the value specified; otherwise retain π exactly or use a suitable approximation.
- ☐ **Correct Final Units Attached:** Area in cm$^\mathbf{2}$/m$^\mathbf{2}$, Volume in cm$^\mathbf{3}$/m$^\mathbf{3}$/L, Mass in g / kg!

TUITION CLASSES